Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Explain what a potential difference of 5 V means in joules per coulomb.

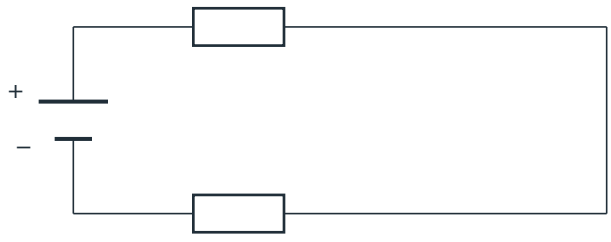
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$.
Series resistors share one unbranched current path.

Worked reasoning: Two series resistors 2 and 4 ohms on a 12 V ideal supply have total 6 ohms and current 2 A. Drops are 4 V and 8 V; they add to 12 V.

Your turn: Draw the loop and label current and both voltage drops.

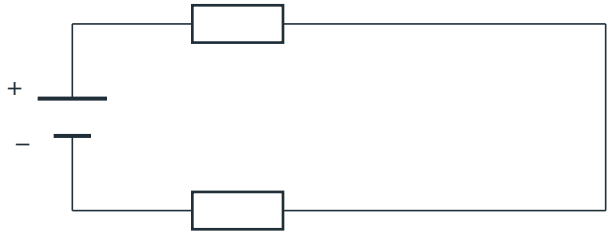


Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Trace the only current path in the series diagram. If either resistor connection opens, what happens to the current?

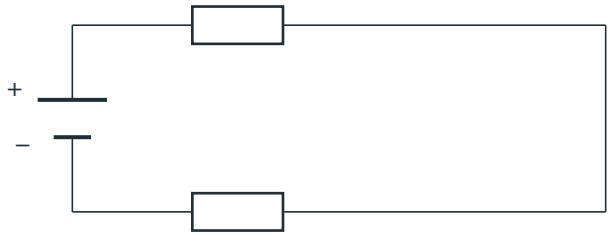


Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 6 V battery powers 2 Ω and 4 Ω resistors in series on a circuit board. Label both resistors and the battery. Add resistances, then find the current through each resistor.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

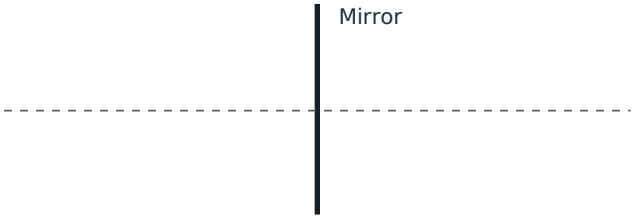
05 Explain your reasoning

A student assigns more current to the smaller series resistor. Explain why this would imply charge accumulating between resistors.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A light ray strikes a plane mirror at 25° from the normal. Find the reflection angle and label both angles on a sketch.
Draw the rays and label both angles from the normal.

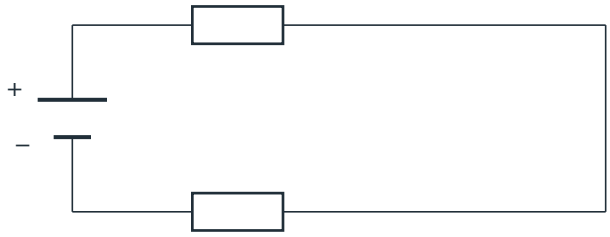


Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

07 Exit check

In a 9 V series loop, one component drops 3 V. Find the other drop if wires are ideal. Label the circuit and show the current path or paths.

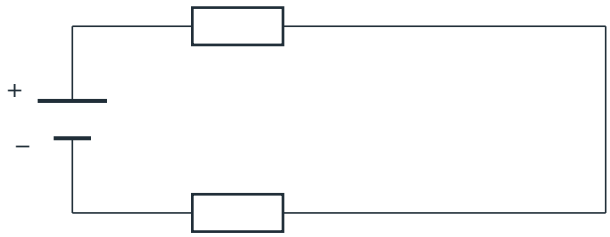


Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose two positive series resistances totaling 10 ohms. On a 10 V supply, find current and each voltage drop. Give a second valid design. Label the circuit and show the current path or paths.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3